SRX400 MNHA Secure Branch Design

GeoIP (ATP Cloud SecIntel), IPsec to a hub, DHCP with Active Leasequery, and Juniper Mist

Eric Sandoval, Network Security Architect, HPE · Release 1.2, November 2026 · SRX400 pair, Junos OS Evolved 25.4X300-D30.3, managed by Juniper Mist (one Hub Profile per node)

Summary. This paper describes a branch design on an **SRX400 Multinode High Availability (MNHA) active/backup pair** managed by Juniper Mist. It covers ATP Cloud enrollment and feed download on both nodes, a **GeoIP** deny policy, per-tenant **source NAT**, an **IPsec overlay to a remote vSRX hub** from each node, and a **DHCP server on the SRG1-active node with Active Leasequery**, which copies the bindings to the peer. Traffic to **Germany (DE)** is denied and traffic to the **United States (US)** permitted on whichever node is active, before and after an SRG1 failover. Traffic to the hub site crosses IPsec from the active node, and the hub's return route follows the active node. A DHCP client keeps its address across a manual failover and failback. Section 2 summarizes the behavior of the design, section 6 gives the expected behavior in detail, and section 9 adds IPv6 on the ISP 1 tenants of node 1.

Mist design rule: configure each SRX400 MNHA node in its **own Hub Profile** (Organization → WAN → Hub Profiles), assigned to that one device — **not in a WAN Edge Template**. Each node's values (MNHA IDs, addresses, SRG1 priority) then sit in its own profile, where the Mist UI shows and edits them directly. Both SRX400 nodes remain overlay **spokes**; the profile name does not change their role. See section 4.2.

Contents

1. Topology

2. Design behavior

3. How it works
- 3.1 ATP Cloud enrollment
- 3.2 GeoIP feed and policy
- 3.3 MNHA: both nodes stay current
- 3.4 Source NAT
- 3.5 Management and cloud services path
- 3.6 DHCP server with MNHA lease synchronization
- 3.7 IPsec to the hub: BGP over IPsec
- 3.8 ISP failure behavior
- 3.9 LAN link failure behavior

4. Mist-managed deployment
- 4.1 What Mist manages
- 4.2 Use Hub Profiles, not a WAN Edge Template
- 4.3 Hub Profile settings (node 2)
- 4.4 Additional CLI: MNHA block (node 2)
- 4.5 Hub Profile for the vSRX hub

5. Configuration
- 5.1 GeoIP object and policy (both nodes)
- 5.2 ATP Cloud / SecIntel (written by enrollment on each node)
- 5.3 Source NAT (egress node)
- 5.4 Cross-node routing and policy (both nodes)
- 5.5 DHCP server with Active Leasequery (node 2 shown)
- 5.6 IPsec to the hub (node 2 shown)

6. Expected behavior
- 6.1 GeoIP before and after a failover
- 6.2 Enrollment, feeds and DHCP
- 6.3 IPsec to the hub before and after a failover
- 6.4 ISP 1 failure on the active node
- 6.5 LAN link failure on the active node

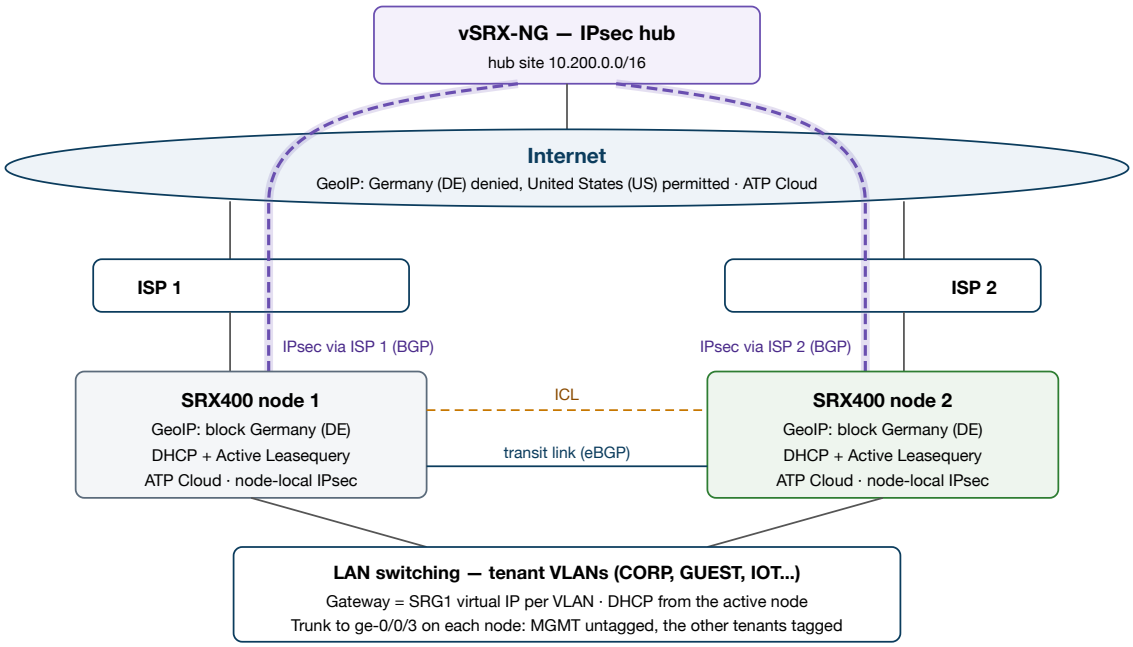
7. Verification checklist

8. Deployment order

9. IPv6 on the ISP 1 tenants (node 1)
- 9.1 How it works
- 9.2 Configuration (node 1 Additional CLI)
- 9.3 Expected behavior

10. Reference configurations

1. Topology



The SRG1-active node owns the tenant gateway virtual IPs and answers DHCP. Each tenant exits through its preferred ISP; a tenant whose ISP is on the peer node crosses the transit link (ge-0/0/5, zone `ICD`). Both nodes carry the GeoIP object and the IOT deny policy on both exit paths (IOT → WAN and IOT → ICD), so whichever node is active has the deny on its exit path (section 6.1). Each node also runs its own IPsec tunnel to the vSRX hub; IOT traffic to the hub site is carried over the tunnel of the active node.

2. Design behavior

Behavior	Sections
Traffic to Germany (DE) is denied by the GeoIP policy and traffic to the United States (US) permitted on whichever node is SRG1-active, before and after a failover	3.2, 5.1, 6.1
IPsec tunnel and BGP to the vSRX hub up on both nodes, whatever their SRG1 role	3.7, 5.6, 6.3
Traffic to the hub site crosses IPsec from whichever node is SRG1-active	3.7, 6.3
The hub's return route follows the active node within about one second	3.7, 6.3
ISP 1 failure on the active node: SRG1 stays; new flows to the hub site continue through the peer's tunnel across the transit link, with about 2.5 s of interruption	3.8, 6.4
LAN link failure on the active node: SRG1 moves to the peer (interface monitoring) and the hub's return path follows it; failback is manual	3.9, 6.5
Each node is enrolled in ATP Cloud by its chassis serial	3.1, 6.2
The GeoIP feed is downloaded on both nodes; GEO-DE holds tens of thousands of IPv4 and IPv6 entries	3.2, 6.2
The backup node keeps refreshing feeds, including after a failover and a failback	3.3, 6.2

Behavior	Sections
Active Leasequery copies the DHCP bindings to the peer; a client keeps its address across a manual failover and failback	3.6, 5.5, 6.2
IPv6 on the ISP 1 tenants of node 1: DHCPv6 prefix delegation on the WAN, one /64 per tenant	9

Platform. SRX400 pair, Junos OS Evolved 25.4X300-D30.3, managed by Juniper Mist (one Hub Profile per node).

All names and addresses in this paper are examples; replace them with your own (section 10).

3. How it works

3.1 ATP Cloud enrollment

Each node is enrolled on its own with the command generated by the ATP Cloud portal (Devices → Enroll), run on the device. The SRX400 is identified and entitled by its **chassis serial**; no license key is installed on the box. Enroll both nodes at the premium tier; enrollment ends with "*Device enrolled successfully!*".

```
request services advanced-anti-malware enroll https://<region>.sky.junipersecurity.net/<enroll-path>/<token>.slax
```

Copy the full enrollment command from the ATP Cloud portal.

3.2 GeoIP feed and policy

The GeoIP category is downloaded once a `dynamic-address` object references it. On each node the object for Germany (DE) holds the feed's DE entries, tens of thousands of IPv4 and IPv6 prefixes. A deny policy that matches the object, placed above the tenant's Internet permit, blocks the country. In an MNHA branch, add it to every zone pair a tenant can exit through: tenant → WAN and tenant → ICD.

Feed state. Before the GeoIP feed has downloaded (for example before ATP enrollment), the GEO-DE object has no entries. Check both the entry count and the GEO-BLOCK-DE hit count (section 7).

3.3 MNHA: both nodes stay current

Each node holds its own ATP Cloud connection and downloads feeds independently, so GeoIP data is already current on the backup when it takes over. The backup node keeps refreshing feeds, including after a manual failover and after failback.

3.4 Source NAT

Each tenant zone source-NATs on the WAN of the node that egresses its traffic. Traffic that arrives from the peer over the inter-node transit link (zone `ICD`, ge-0/0/5) is translated by the egress node's `ICD_to_wan` (node 1) or `ICD_to_wan2` (node 2) rule-set. Both nodes use interface NAT (section 5.3).

The SRX400 supports one data SRG (SRG1), so a tenant whose ISP is on the peer node crosses the transit link in steady state, not only during failures. The GeoIP deny sits on both exit paths of IOT, IOT → WAN and IOT → ICD (section 5.1), so traffic that crosses the transit link meets the same policy.

3.5 Management and cloud services path

The design separates management from tenant traffic. inet.0's only default route is the one to the out-of-band management gateway (`qualified-next-hop <mgmt-gateway> preference 1`), and the WAN port (ge-0/0/2) and the transit link (ge-0/0/5) sit in routing instance `WAN-VR` (section 5.4). The Mist cloud connection, the ATP Cloud and SecIntel connections and DNS therefore leave each node by its management port (`re0:mgmt-0` on the SRX400), and tenant traffic leaves by `WAN-VR`. Give each node's management network a path to Mist, ATP Cloud and DNS.

3.6 DHCP server with MNHA lease synchronization

`managed-services dhcp` makes the SRG1-active node the DHCP server. On its own it does not copy bindings to the peer. **Active Leasequery** (ALQ, RFC 7724) does: it copies the bindings to the peer over a TCP session (TCP/67) between the nodes.

- **One ALQ peer per DHCP routing instance.** The peer address must be in the same routing instance as the DHCP group and pool. This design has two DHCP routing instances: `lan` (MGMT, CORP, MEDIA, SECURITY, IOT) and `GUEST-VR` (GUEST). Each node peers with the other node's address in each instance: MGMT 10.10.10.2 / .3 in `lan`, GUEST 10.10.30.2 / .3 in `GUEST-VR`.
- **The switch must carry each peer VLAN between the nodes.** That means MGMT (VLAN 10, untagged on the LAN trunk) for `lan` and GUEST (VLAN 30) for `GUEST-VR`.
- **Broadcast replies in `lan`.** The `lan` DHCP server sends OFFER and ACK as broadcast (section 5.5).
- **Failback is manual** (preemption disabled). On the active node, run `request chassis high-availability failover services-redundancy-group 1 peer-id <target>`. First check on the target node: cold sync COMPLETE in `show chassis high-availability information`, and the ALQ peer Connected in both `lan` and `GUEST-VR` (section 7).

Behavior. After a manual failover, and again after failback, the client keeps the **same IP**. A release removes the binding from both nodes within about 2 s.

3.7 IPsec to the hub: BGP over IPsec

Each SRX400 node builds its own IKEv2 tunnel to the remote vSRX hub from its own WAN: certificate (RSA signature) authentication, DH group 14, AES-256-GCM for IKE and ESP, dead-peer detection every 3 s. Tunnels are `node-local`, the MNHA mode for per-node tunnels with dynamic routing, so each node keeps its own tunnel up whatever its SRG1 role, including on the backup. Each node runs iBGP to the hub over its tunnel with BFD (1 s × 3) and advertises the tenant subnet; the SRG1-active node adds local-preference 200, using the active signal route as a policy condition. The hub therefore returns traffic to the active node, and moves to the new active node as soon as SRG1 fails over.

What is synchronized. Firewall sessions are synchronized between the nodes; IPsec security associations are not. Each node negotiates its own tunnel (`node-local`), so after a failover new flows use the new active node's tunnel, steered by the BGP local-preference.

3.8 ISP failure behavior

When the active node loses its ISP, SRG1 stays on that node (the LAN is unaffected). The node no longer has its own tunnel, so it forwards the tenant's hub-site traffic across the transit link (ge-0/0/5) to the peer, which sends it out its own tunnel; the hub returns traffic to the peer, which hands it back over the transit link. When the ISP returns, the tunnel re-establishes and the hub moves back to the primary path automatically (section 6.4).

3.9 LAN link failure behavior

SRG1 monitors the LAN interface (`monitor interface ge-0/0/3`). If the active node loses its LAN link, it goes INELIGIBLE (failure event `IF`) and the peer takes over SRG1: the tenant virtual IPs move, and the active signal route moves with them, so the hub's return path follows the new active node (local-preference 200). When the link returns, the node rejoins as BACKUP. With preemption disabled, moving SRG1 back is a manual command (section 6.5).

4. Mist-managed deployment

4.1 What Mist manages

The SRX400 pair is managed by Juniper Mist. Each node has its own Hub Profile (section 4.2), so the two nodes receive their own configuration. Mist renders the ports, tenant networks, DHCP, application policies and the IPsec overlay natively. Everything that makes the two nodes work as a pair is carried in each profile's Additional CLI: MNHA, the routing and policies between the nodes, the cross-node NAT, the overlay settings Mist does not render, DHCP lease synchronization and GeoIP.

Rendered natively by Mist	Carried in Additional CLI (per node profile)
• WAN and LAN ports, tenant networks and VLANs, and the node's own address on each network • Tenant networks in routing instance `lan` • DHCP servers and pools • From the application policies and each network's Internet access: tenant → own WAN zone and IOT → overlay security policies, and per-tenant source NAT on the node's own WAN • IPsec overlay to the vSRX hub: certificate-based IKEv2, multipoint `st0`, BGP to the hub (instance `vpn_OrgOverlay`)	• MNHA: SRG1, ICL, MNHA forwarding over the ICL, virtual IPs, signal routes, LAN interface monitoring (section 4.4) • VR routing: `WAN-VR` holding the WAN (ge-0/0/2) and the transit link (ge-0/0/5), the `lan` default next-table WAN-VR, `GUEST-VR` for GUEST and its DHCP; Mist APBR removed (`delete apply-groups apbr`) (section 5.4) • Inter-node transit link ge-0/0/5: eBGP between the nodes with BFD, policies `WAN-EXPORT`, `ICD-IMPORT` and `HUB-FROM-OVERLAY`, rib-group `WAN-LEAK` • Zone `ICD` and the cross-node policies: tenant → ICD (`*-xnode`), ICD → WAN (`xnode-out`), ICD → vpn_OrgOverlay (`xnode-iot`) • ICD source NAT: interface NAT, rule-set `ICD_to_wan` on node 1 and `ICD_to_wan2` on node 2 (section 5.3) • Overlay: AS 65000 in vpn_OrgOverlay, `node-local` on the IKE gateway, BGP BFD, active-node local-preference (`IOT-LP` with the `MNHA-ACTIVE` condition), DPD always-send on the Mist-generated IKE gateway (section 5.6) • DHCP `managed-services dhcp` and Active Leasequery, one peer per DHCP routing instance (lan and GUEST-VR), and broadcast replies in lan (section 5.5) • GeoIP `dynamic-address` and the GEO-BLOCK-DE deny policies (IOT → WAN and IOT → ICD) • ATP Cloud / SecIntel connection lines (section 5.2) • Management separation: the inet.0 default to the out-of-band management gateway (section 3.5)

- **ATP Cloud enrollment** is done on each device with the ATP Cloud portal command (section 3.1). Enrollment writes PKI, SSL and ATP Cloud configuration on the box.
- **What a Mist push keeps.** Mist pushes the full profile on every change. Configuration written by enrollment stays in place across Mist pushes. Apart from what enrollment writes, keep every change in the profile's Additional CLI.

- **Removed `chassis high-availability` lines stay on the box.** Removing such a line from the Additional CLI stops Mist sending it, but the statement stays in the running configuration of each node until it is deleted on the box. After removing any Additional CLI line, check the box for it.
- **Enrollment lines in the profile.** The profiles also carry the section 5.2 lines, which match what enrollment writes. Those lines reference CA profiles and a certificate that only enrollment creates.
- **Set, delete and comment lines only.** Additional CLI accepts `set`, `delete` and `#` comment lines (Figure 4-5). An `insert` line fails the whole push, and Mist shows no reason. Policy order follows line order (section 5.1).
- **Confirm on the box.** The Mist portal status is not final: a commit shown as FAILED can later show COMMITTED, for example after a retry. After each profile change, check `show system commit` for the Mist commit.

4.2 Use Hub Profiles, not a WAN Edge Template

> **Build each SRX400 node in its own Hub Profile** (Organization → WAN → Hub Profiles), assigned to that one device. **Do not use a WAN Edge Template** for the MNHA pair.

The two MNHA nodes must differ: each has its own LAN addresses, ICL and inter-node transit link addresses, MNHA local and peer IDs and SRG1 priority. A WAN Edge Template is shared by the site's gateways. With MNHA enabled, Mist keeps each node's values in node rules (node 0, node 1) inside the template, and some per-node values in those rules can be set only through the Mist API. This design keeps each node's values in its own Hub Profile, assigned to that device. The Mist UI shows and edits them in the profile (Figures 4-3 to 4-5). Despite the name, a Hub Profile does not make the SRX400 an overlay hub — the overlay role is set per WAN port, and both SRX400 nodes are **spokes**. The vSRX hub uses a third Hub Profile with the `hub` role.

Hub Profile	Assigned to	Role
SRX400-1 (node 1 profile)	SRX400 node 1	Spoke on the overlay; MNHA local-id 1, SRG1 priority 200
SRX400-2 (node 2 profile)	SRX400 node 2	Spoke on the overlay; MNHA local-id 2, SRG1 priority 100
HUB (vSRX hub profile)	vSRX-NG	Overlay hub (static WAN address, `role: hub`)

Device-level Additional CLI. Mist also accepts an Additional CLI list on the device itself. A device-level list replaces the profile's list; the two are not merged. If a node has one, keep it identical to the profile's list: an edit made only in the profile does not reach that node. Never set the device-level list to an empty list (`[]`) or a partial one: Mist then renders the node from that list alone. An empty device-level list removes the MNHA support lines from the node (`allow-ha`, the cross-node NAT, `MNHA-NODE-LOCAL`, the WAN port's WAN-VR membership, the management default), while its `chassis high-availability` lines stay (section 4.1). The node loses its Mist connection, and both nodes go ACTIVE until Junos rolls back the Mist commit (commit confirmed, about 5 minutes).

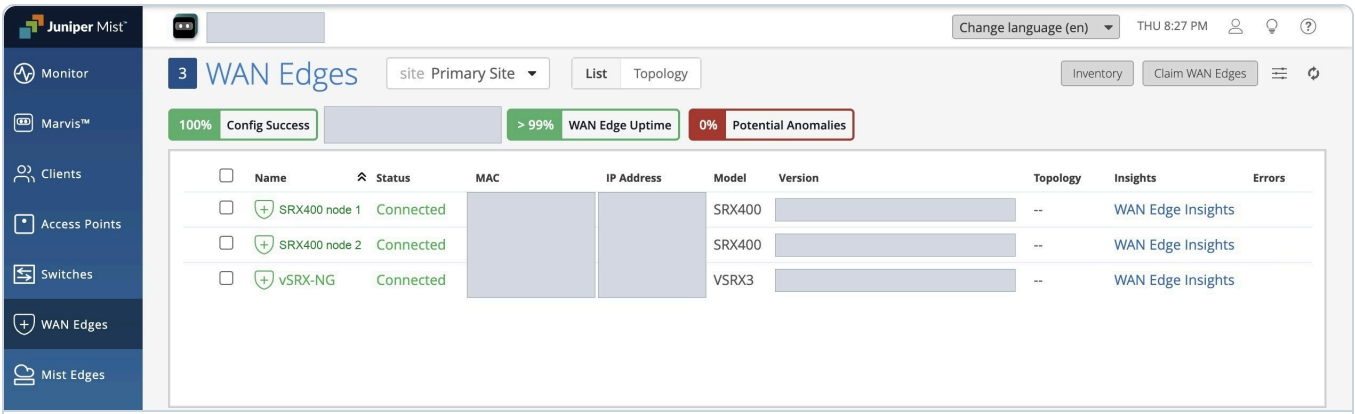


Figure 4-1: WAN Edges inventory: both SRX400 nodes and the vSRX hub connected

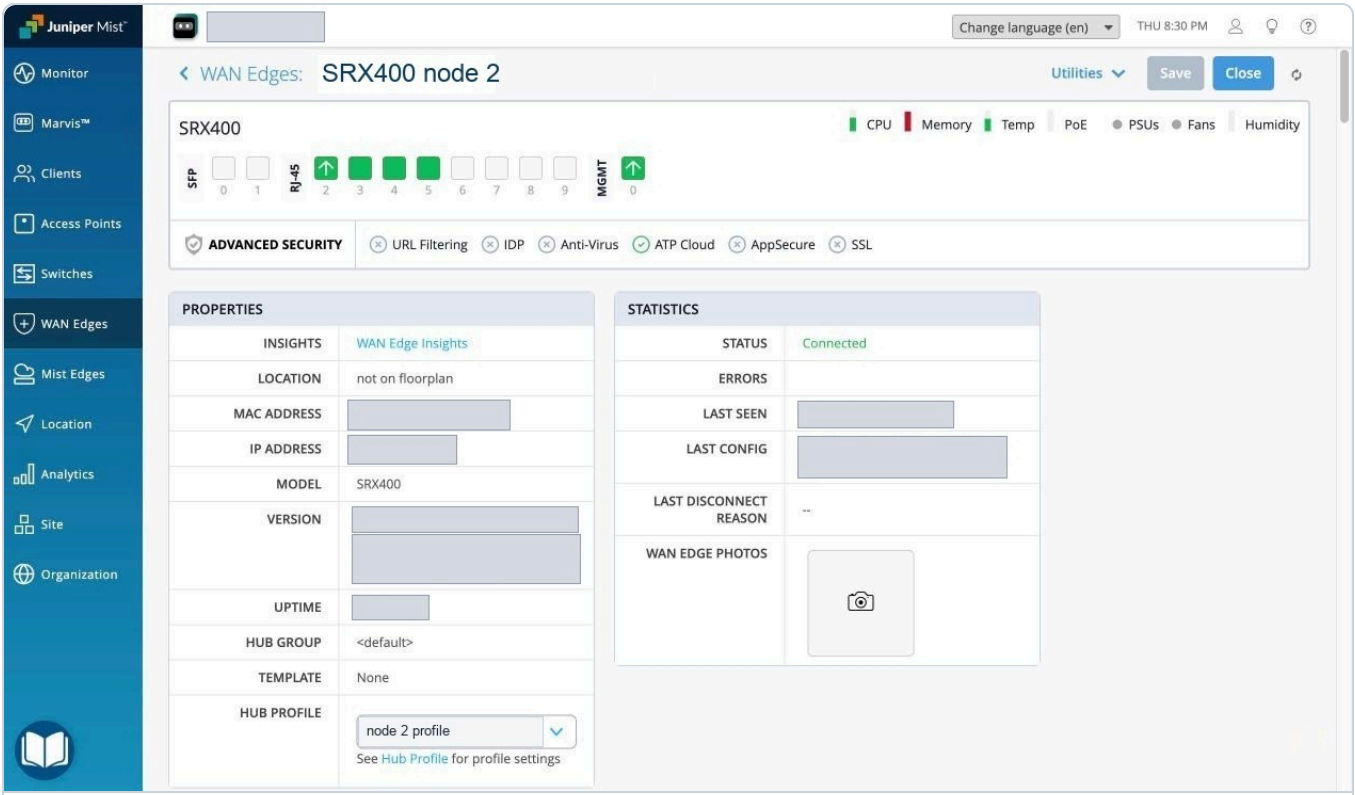


Figure 4-2: SRX400 device page: port status, ATP Cloud enabled under Advanced Security, and the node's own Hub Profile

4.3 Hub Profile settings (node 2)

These are the native settings in node 2's Hub Profile. From them Mist renders the ports, networks, DHCP, security policies, source NAT and the overlay. The Additional CLI then removes Mist's APBR (`delete apply-groups apbr`), so the paths do not steer traffic. VR routing decides which ISP each tenant uses (second table below).

Mist setting	Node 2 value	What it does
WAN port	ge-0/0/2, name **wan2**, broadband, DHCP	Internet uplink to ISP 2. The Additional CLI places ge-0/0/2 in WAN-VR (section 5.4)
WAN overlay (Hub to Hub endpoint)	Spoke on the hub's overlay path, BFD profile broadband	Builds the IPsec tunnel to the vSRX hub. The BFD profile is the path setting; BFD on the overlay BGP (1000 ms × 3) is Additional CLI

Mist setting	Node 2 value	What it does
LAN port	ge-0/0/3: untagged MGMT (VLAN 10); tagged CORP 20, GUEST 30, MEDIA 40, SECURITY 50, IOT 60	Tenant VLAN trunk to the LAN switch
IP config per network	MGMT 10.10.10.**3**/24, CORP 10.10.20.**3**/24, IOT 10.10.60.**3**/24, ... (node 1 uses .2)	The node's own address on each tenant network; the gateway (.1) stays the SRG1 virtual IP
DHCP config	Server on each tenant network, range .10–.200, router .1	Address pools handed out by the active node. The Additional CLI moves the GUEST server and pool to GUEST-VR
VPN access (IOT)	OrgOverlay: routed, no re-advertise to overlay	Advertises the IOT subnet to the hub over the overlay BGP
Traffic steering	Path **wan** = port wan2 · path **OrgOverlay** = the hub overlay path	Named paths the application policies below use. APBR is removed; VR routing decides egress
Application policy **allow-to-hub**	IOT → HUB-SITE-NETS (hub site 10.200.0.0/16), path OrgOverlay	Permits IOT → vpn_OrgOverlay; IOT traffic to the hub site rides IPsec
Application policy **Internet**	All six tenant networks → any, path wan	Permits each tenant → wan2 (`01_Internet`) with source NAT on wan2. VR routing, not this path, sets the ISP a tenant uses

Egress by VR routing (Additional CLI). Each tenant routing instance has a default route `0.0.0.0/0 next-table WAN-VR.inet.0`. Each node also learns its peer's ISP default over ge-0/0/5 (BGP, preference 170), and rib-group WAN-LEAK copies it into lan and GUEST-VR. Where the local default has preference 200, the peer's default wins and the tenant crosses to the peer (section 5.4).

Routing instance	Tenants	Node 1	Node 2
`lan` — ISP 1 side	MGMT, CORP, MEDIA, SECURITY, IOT	Own WAN (ISP 1): local next-table default	Peer's ISP 1 across ge-0/0/5: BGP default (170) beats the local default (preference 200)
`GUEST-VR` — ISP 2 side	GUEST	Peer's ISP 2 across ge-0/0/5: BGP default (170) beats the local default (preference 200)	Own WAN (ISP 2): local next-table default

On the egress node, traffic that crosses ge-0/0/5 is permitted by the Additional CLI policies `xnode-out` (ICD → WAN) and `xnode-iot` (ICD → vpn_OrgOverlay), not by the application policies above. The `ICD_to_wan` (node 1) or `ICD_to_wan2` (node 2) rule-set translates the traffic that leaves by the WAN (interface NAT). VR lan learns the hub-site prefixes from the overlay BGP, so IOT traffic to the hub uses the active node's tunnel. The preference-200 default is the fallback to the local ISP when the peer's default is absent. WAN-VR keeps the WAN out of inet.0, whose only default route is the out-of-band management default (section 3.5).

Node 1 is identical except for its WAN port name (**wan**, so path wan = port wan) and its own addresses (.2 on each tenant network).

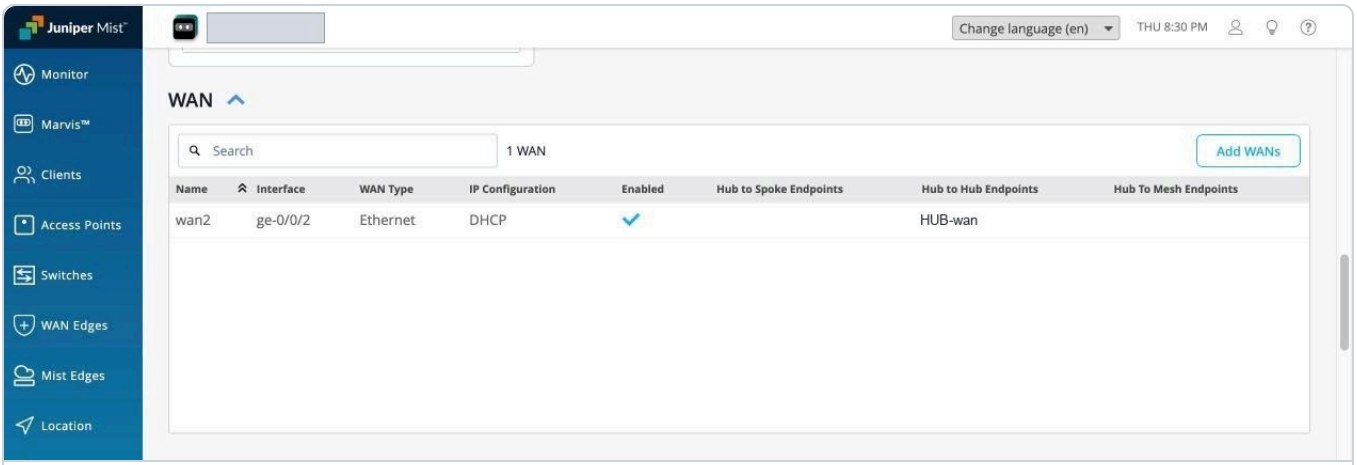


Figure 4-3: Hub Profile: WAN port with its overlay path to the hub

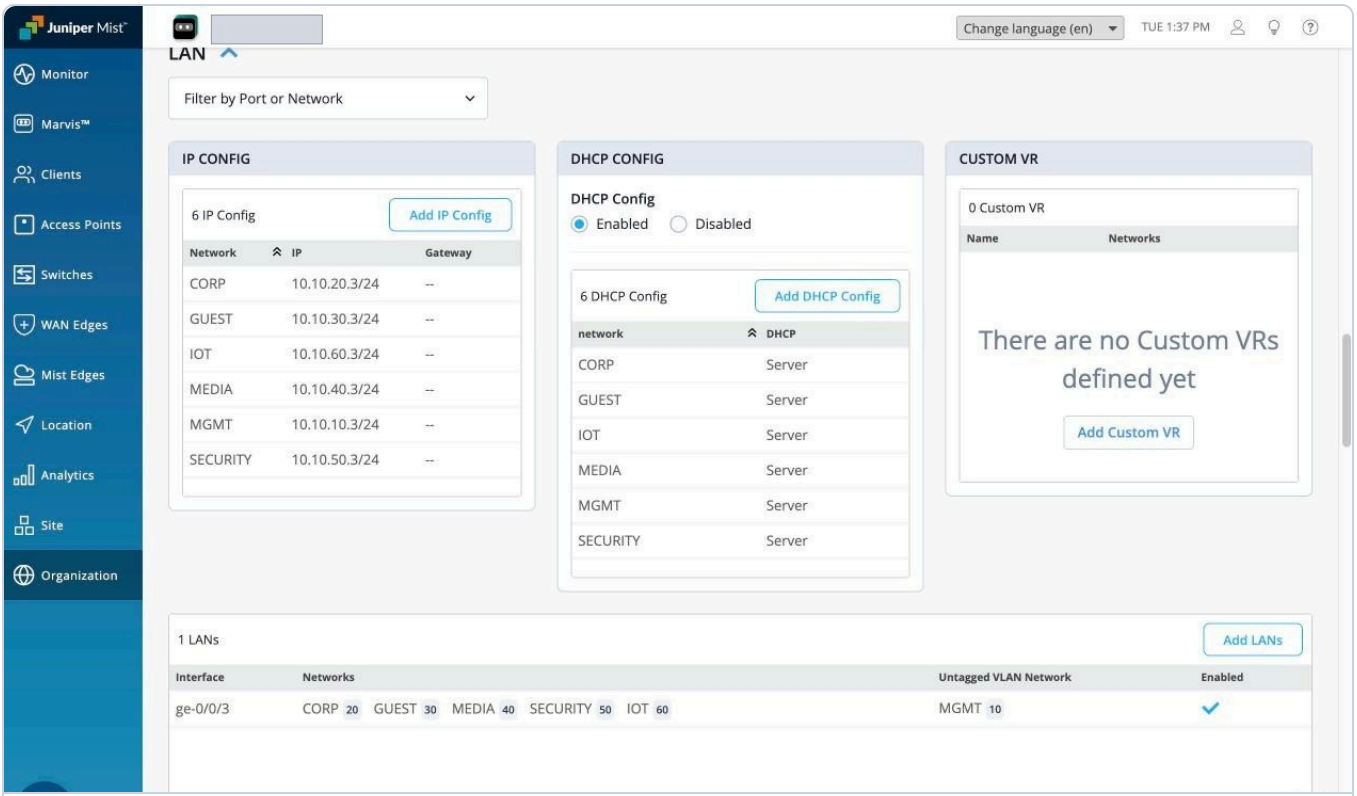


Figure 4-4: Hub Profile: per-node IP configuration on each tenant network, DHCP servers, and the LAN trunk

4.4 Additional CLI: MNHA block (node 2)

This design carries MNHA in each node's Additional CLI: the services redundancy group, the ICL, the MNHA forwarding path and the virtual IPs. Below is node 2's full MNHA block from its profile. The routing, policies and NAT between the nodes are separate Additional CLI items (section 4.1, section 5.4).

```
# node identity and ICL (ge-0/0/4)
set chassis high-availability local-id 2
set chassis high-availability local-id local-ip 10.255.0.2
set chassis high-availability local-id local-forwarding-ip 10.255.1.2
set chassis high-availability peer-id 1 peer-ip 10.255.0.1
set chassis high-availability peer-id 1 interface ge-0/0/4.0
set chassis high-availability peer-id 1 liveness-detection minimum-interval 1000
set chassis high-availability peer-id 1 liveness-detection multiplier 3
set interfaces ge-0/0/4 unit 0 family inet address 10.255.0.2/30
```

```
# MNHA forwarding path on the ICL link
set chassis high-availability peer-id 1 peer-forwarding-ip 10.255.1.1
set chassis high-availability peer-id 1 peer-forwarding-ip interface ge-0/0/4.0
set chassis high-availability peer-id 1 peer-forwarding-ip liveness-detection minimum-interval 1000
set chassis high-availability peer-id 1 peer-forwarding-ip liveness-detection multiplier 3
set interfaces lo0 unit 0 family inet address 10.255.1.2/32
set routing-options static route 10.255.1.1/32 next-hop 10.255.0.1

# zone HA for the ICL and the forwarding addresses
set security zones security-zone HA interfaces ge-0/0/4.0 host-inbound-traffic system-services all
set security zones security-zone HA interfaces ge-0/0/4.0 host-inbound-traffic protocols all
set security zones security-zone HA interfaces lo0.0 host-inbound-traffic system-services all
set security zones security-zone HA interfaces lo0.0 host-inbound-traffic protocols all
set security policies from-zone HA to-zone HA policy allow-ha match source-address any
set security policies from-zone HA to-zone HA policy allow-ha match destination-address any
set security policies from-zone HA to-zone HA policy allow-ha match application any
set security policies from-zone HA to-zone HA policy allow-ha then permit

# SRG1: hybrid, LAN interface monitoring, signal routes, DHCP
set chassis high-availability services-redundancy-group 1 deployment-type hybrid
set chassis high-availability services-redundancy-group 1 peer-id 1
set chassis high-availability services-redundancy-group 1 activeness-priority 100
set chassis high-availability services-redundancy-group 1 monitor interface ge-0/0/3
set chassis high-availability services-redundancy-group 1 active-signal-route 10.255.3.1
set chassis high-availability services-redundancy-group 1 backup-signal-route 10.255.3.2
set chassis high-availability services-redundancy-group 1 managed-services dhcp

# SRG1 virtual IPs: the tenant gateways (.1) on the LAN units of ge-0/0/3
set chassis high-availability services-redundancy-group 1 virtual-ip 1 ip 10.10.10.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 1 interface ge-0/0/3.0
set chassis high-availability services-redundancy-group 1 virtual-ip 1 use-virtual-mac
set chassis high-availability services-redundancy-group 1 virtual-ip 2 ip 10.10.20.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 2 interface ge-0/0/3.20
set chassis high-availability services-redundancy-group 1 virtual-ip 2 use-virtual-mac
set chassis high-availability services-redundancy-group 1 virtual-ip 3 ip 10.10.30.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 3 interface ge-0/0/3.30
set chassis high-availability services-redundancy-group 1 virtual-ip 3 use-virtual-mac
set chassis high-availability services-redundancy-group 1 virtual-ip 4 ip 10.10.40.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 4 interface ge-0/0/3.40
set chassis high-availability services-redundancy-group 1 virtual-ip 4 use-virtual-mac
set chassis high-availability services-redundancy-group 1 virtual-ip 5 ip 10.10.50.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 5 interface ge-0/0/3.50
set chassis high-availability services-redundancy-group 1 virtual-ip 5 use-virtual-mac
set chassis high-availability services-redundancy-group 1 virtual-ip 6 ip 10.10.60.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 6 interface ge-0/0/3.60
set chassis high-availability services-redundancy-group 1 virtual-ip 6 use-virtual-mac

# active signal route as a policy condition (overlay local-preference, ge-0/0/5 export)
set policy-options condition MNHA-ACTIVE if-route-exists address-family inet 10.255.3.1/32
set policy-options condition MNHA-ACTIVE if-route-exists address-family inet table inet.0

# node-local on the IKE gateway Mist generates for the overlay
set groups MNHA-NODE-LOCAL security ike gateway <*> node-local
set apply-groups MNHA-NODE-LOCAL

# inter-node transit link (eBGP) - not part of MNHA, see the note below
set interfaces ge-0/0/5 unit 0 family inet address 10.255.2.2/30
```

ICL and MNHA forwarding. ge-0/0/4 is one link. It carries the ICL (10.255.0.x/30), session synchronization and the MNHA forwarding path. `peer-forwarding-ip interface ge-0/0/4.0` puts the forwarding path on the ICL. The forwarding IPs sit on lo0.0, and a static route over the ICL reaches the peer's.

Inter-node transit link (eBGP). ge-0/0/5 (10.255.2.x/30) is a routed link in WAN-VR. It carries tenant traffic whose ISP is on the peer node, hub-site traffic during an ISP failure, and the eBGP session between the nodes (AS 65001 / 65002, BFD 300 ms × 3; section 5.4). MNHA does not track it. Its security zone and BGP group are named `ICD`; the NAT rule-sets are `ICD_to_wan` and `ICD_to_wan2`. It is not the MNHA ICD; MNHA forwarding uses the ICL.

No activeness-probe and no preemption are configured. Moving SRG1 back after a failover is a manual command. A wildcard group applies `node-local` to the IKE gateway Mist generates for the overlay, whatever its generated name.

Node 1 differences. Everything else in the block is identical, including the virtual IPs and signal routes.

- `local-id 1`, `local-ip 10.255.0.1`, `local-forwarding-ip 10.255.1.1`
- `peer-id 2` in every peer-id line and in SRG1; `peer-ip 10.255.0.2`, `peer-forwarding-ip 10.255.1.2`

- SRG1 `activeness-priority 200`
- ge-0/0/4 `10.255.0.1/30`, lo0.0 `10.255.1.1/32`, ge-0/0/5 `10.255.2.1/30`
- `set routing-options static route 10.255.1.2/32 next-hop 10.255.0.2`

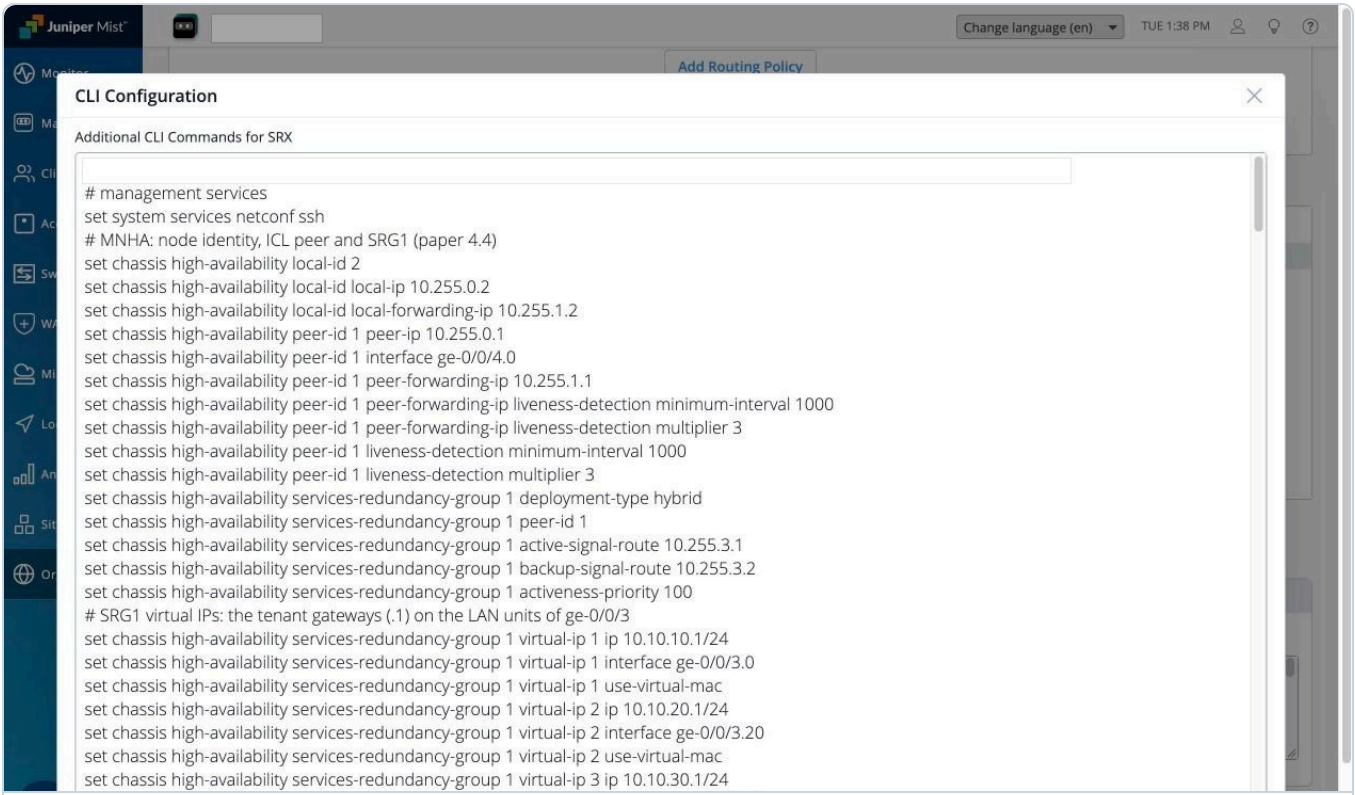


Figure 4-5: Hub Profile: Additional CLI carrying the MNHA statements

4.5 Hub Profile for the vSRX hub

The vSRX hub has its own Hub Profile with the `hub` role.

Mist setting	Hub value	What it does
WAN port	ge-0/0/0, static 172.16.11.2/30, gateway 172.16.11.1	Fixed address the spokes connect to
WAN overlay	Hub role on the overlay path, BFD profile broadband	Terminates both SRX400 tunnels

Mist builds the hub side of the overlay from this profile: certificate-based IKEv2, multipoint `st0` interfaces (four for the one hub path: `st0.0` to `st0.3`, each with its own IPsec VPN on the one hub IKE gateway), and the iBGP route reflector toward the spokes in routing instance `vpn_OrgOverlay` (BGP group `OrgOverlay_overlay`, export `OrgOverlay_advertise`). Mist's rendered export matches BGP and OSPF routes only, so it does not advertise the hub's static routes. The hub's Additional CLI adds the hub-site advertisement, an import filter, BGP BFD and DPD always-send.

```
# hub-site prefix behind the hub, advertised to the spokes (discard route; hub-site address on lo0.1)
set routing-instances vpn_OrgOverlay routing-options static route 10.200.0.0/16 discard
set policy-options policy-statement OrgOverlay_advertise term 10_hub_site from protocol static
set policy-options policy-statement OrgOverlay_advertise term 10_hub_site from route-filter 10.200.0.0/16 exact
set policy-options policy-statement OrgOverlay_advertise term 10_hub_site then next-hop self
set policy-options policy-statement OrgOverlay_advertise term 10_hub_site then accept

# hub-site address on the overlay loopback (reachability target, sections 6.3 and 8)
set interfaces lo0 unit 1 family inet address 10.200.8.1/32

# spokes must not leak MNHA control addresses into the overlay
set policy-options policy-statement OVERLAY-IN term no-mnha from route-filter 10.255.0.0/16 orlonger
set policy-options policy-statement OVERLAY-IN term no-mnha then reject
set policy-options policy-statement OVERLAY-IN term rest then accept
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay import OVERLAY-IN

# BFD on the overlay iBGP, same timers as the spokes (BGP hold time is 90 s)
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay bfd-liveness-detection minimum-interval 1000
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay bfd-liveness-detection multiplier 3

# DPD always-send on the Mist-generated hub gateway (Mist renders optimized)
delete security ike gateway OrgOverlay_hub_on_wan dead-peer-detection optimized
set security ike gateway OrgOverlay_hub_on_wan dead-peer-detection always-send
```

In this design the hub site is one prefix, 10.200.0.0/16 (a discard route on the hub), and the hub-site address 10.200.8.1 on `lo0.1` is the reachability target of section 6.3 and of section 8, step 10. Replace them with the hub site's own prefixes. The `10_hub_site` term exports static routes only. `OVERLAY-IN` rejects only 10.255.0.0/16 and accepts every other spoke route. Mist generates the gateway name `OrgOverlay_hub_on_wan`; take it from the hub's rendered configuration.

The hub's own management and Internet access are installation-specific and are not part of this design.

5. Configuration

5.1 GeoIP object and policy (both nodes)

These lines are in each node's Hub Profile Additional CLI. Policies in a zone pair are evaluated in list order, so the deny must sit above the permit. Mist Additional CLI accepts only `set`, `delete` and `#` comment lines. An `insert` line fails the whole push. The order is therefore set this way:

- **ICD path.** All IOT → ICD policies come from Additional CLI. GEO-BLOCK-DE is listed before IOT-xnode, the cross-node permit, so it is evaluated first. The GEO-BLOCK-DE lines sit just above the IOT-xnode lines of section 5.4 in the list. Put them there; do not append them at the end of the list.
- **WAN path.** Mist renders the IOT Internet permit as `01_Internet`. GEO-BLOCK-DE is set after it. Then `01_Internet` is deleted and set again with Mist's exact rendered lines. The re-set policy goes to the end of the list, after the deny.

```
# GEO-DE = all addresses the GeoIP feed maps to Germany (ISO code DE)
set security dynamic-address address-name GEO-DE profile category GeoIP property country string DE

# ICD path (both nodes): GEO-BLOCK-DE listed before IOT-xnode
set security policies from-zone IOT to-zone ICD policy GEO-BLOCK-DE match source-address any
set security policies from-zone IOT to-zone ICD policy GEO-BLOCK-DE match destination-address GEO-DE
set security policies from-zone IOT to-zone ICD policy GEO-BLOCK-DE match application any
set security policies from-zone IOT to-zone ICD policy GEO-BLOCK-DE then deny
set security policies from-zone IOT to-zone ICD policy GEO-BLOCK-DE then count
set security policies from-zone IOT to-zone ICD policy IOT-xnode match source-address any
set security policies from-zone IOT to-zone ICD policy IOT-xnode match destination-address any
set security policies from-zone IOT to-zone ICD policy IOT-xnode match application any
set security policies from-zone IOT to-zone ICD policy IOT-xnode then permit

# WAN path (node 1 shown; node 2 uses zone wan2)
set security policies from-zone IOT to-zone wan policy GEO-BLOCK-DE match source-address any
set security policies from-zone IOT to-zone wan policy GEO-BLOCK-DE match destination-address GEO-DE
set security policies from-zone IOT to-zone wan policy GEO-BLOCK-DE match application any
set security policies from-zone IOT to-zone wan policy GEO-BLOCK-DE then deny
set security policies from-zone IOT to-zone wan policy GEO-BLOCK-DE then count
# remove Mist's 01_Internet, then re-set it with Mist's exact rendered lines (copy them from the current render)
```

```
delete security policies from-zone IOT to-zone wan policy 01_Internet
set security policies from-zone IOT to-zone wan policy 01_Internet match source-address 10-10-60-0_24
set security policies from-zone IOT to-zone wan policy 01_Internet match destination-address any
set security policies from-zone IOT to-zone wan policy 01_Internet match application any
set security policies from-zone IOT to-zone wan policy 01_Internet match dynamic-application any
set security policies from-zone IOT to-zone wan policy 01_Internet then permit
```

Mist generates the name `01_Internet` and the address name `10-10-60-0_24` from the application policy and the IOT network. Copy the re-set lines from Mist's current render of the device. Check them again after any application-policy change. Check the order on both nodes with `show security policies from-zone IOT to-zone wan | no-more` (`wan2` on node 2) and `show security policies from-zone IOT to-zone ICD | no-more`: GEO-BLOCK-DE has sequence number 1.

The design applies GEO-BLOCK-DE to IOT. The WAN zone name is per node: `wan` on node 1, `wan2` on node 2.

5.2 ATP Cloud / SecIntel (written by enrollment on each node)

```
set services advanced-anti-malware connection url https://srxapi-v2.<region>.sky.junipersecurity.net
set services advanced-anti-malware connection authentication tls-profile aamw-ssl
set services security-intelligence url https://cloudfeeds-v2.sky.junipersecurity.net/api/manifest.xml
set services security-intelligence authentication tls-profile aamw-ssl
set services ssl initiation profile aamw-ssl trusted-ca aamw-secintel-ca
set services ssl initiation profile aamw-ssl trusted-ca aamw-cloud-ca
set services ssl initiation profile aamw-ssl client-certificate aamw-srx-cert
set services ssl initiation profile aamw-ssl actions crl disable
```

5.3 Source NAT (egress node)

Both nodes use interface NAT. Traffic is translated to the address of the WAN interface it leaves by: ge-0/0/2, ISP 1 on node 1 and ISP 2 on node 2. No NAT pool is configured. Mist renders the tenant NAT. The NAT for traffic that arrives from the peer over ge-0/0/5 is in Additional CLI, because zone `ICD` is defined there. The rule-set names differ per node because the WAN zone name differs.

Traffic	Configured by	Node 1	Node 2
Tenant zone → local WAN	Mist: Internet application policy and Internet access on each network	One rule-set per tenant zone, `<tenant>_to_wan`, `source-nat interface`	One rule-set per tenant zone, `<tenant>_to_wan2`, `source-nat interface`
ICD → local WAN (traffic from the peer)	Additional CLI	`ICD_to_wan`, rule `xnode`	`ICD_to_wan2`, rule `xnode`

```
# node 2 (Additional CLI)
set security nat source rule-set ICD_to_wan2 from zone ICD
set security nat source rule-set ICD_to_wan2 to zone wan2
set security nat source rule-set ICD_to_wan2 rule xnode match source-address 0.0.0.0/0
set security nat source rule-set ICD_to_wan2 rule xnode then source-nat interface

# node 1 (Additional CLI)
set security nat source rule-set ICD_to_wan from zone ICD
set security nat source rule-set ICD_to_wan to zone wan
set security nat source rule-set ICD_to_wan rule xnode match source-address 0.0.0.0/0
set security nat source rule-set ICD_to_wan rule xnode then source-nat interface
```

The ICD rule-set needs the `ICD → wan|wan2` permit `xnode-out` (section 5.4). `show security nat source rule all` lists every rule-set with its translation hits.

5.4 Cross-node routing and policy (both nodes)

Routing-instance (VR) routing decides which ISP each tenant uses. Mist renders the tenant networks, the application policies and the tenant NAT. The Additional CLI below does four things. It removes Mist's APBR. It

puts the WAN in `WAN-VR`, which keeps it out of inet.0 (section 3.5). It gives each tenant VR a default into `WAN-VR`. It joins the `WAN-VR` instances of the two nodes with eBGP over ge-0/0/5.

ge-0/0/5 is the inter-node transit link; its zone and BGP group are both named `ICD`. It is a routed link. MNHA forwarding uses the ICL, ge-0/0/4 (section 4.4). Node 2 is shown first, then node 1's differences.

Routing instance (tenants)	Node 1	Node 2
`lan` (MGMT, CORP, MEDIA, SECURITY, IOT)	Static 0/0 next-table `WAN-VR`, default preference (5). Exits ISP 1 locally.	Static 0/0 next-table `WAN-VR`, preference 200. The BGP 0/0 from node 1 (170) is preferred. Exits ISP 1 through node 1.
`GUEST-VR` (GUEST)	Static 0/0 next-table `WAN-VR`, preference 200. The BGP 0/0 from node 2 (170) is preferred. Exits ISP 2 through node 2.	Static 0/0 next-table `WAN-VR`, default preference (5). Exits ISP 2 locally.
`WAN-VR`	DHCP default from ISP 1; BGP 0/0 from node 2	DHCP default from ISP 2; BGP 0/0 from node 1
Hub-site prefixes (10.200.0.0/16)	Each node uses its own tunnel first: `lan` imports these prefixes from `vpn_OrgOverlay` through Mist's `vpn_OrgOverlay_bgp`. `WAN-VR` holds the node's own copy (`HUB-FROM-OVERLAY`, BGP 170) and the peer's copy at preference 175. When a node's own tunnel is down, the `lan` default carries hub traffic across ge-0/0/5 to the peer's tunnel (section 6.4).	

WAN-VR and the transit link (node 2)

```
set interfaces ge-0/0/5 unit 0 family inet address 10.255.2.2/30
# management separation: inet.0 keeps only the out-of-band management default (section 3.5)
set routing-options static route 0.0.0.0/0 qualified-next-hop <mgmt-gateway> preference 1
# the WAN (ge-0/0/2, Mist WAN port, DHCP) sits in WAN-VR with the transit link
set routing-instances WAN-VR instance-type virtual-router
set routing-instances WAN-VR interface ge-0/0/2.0
set routing-instances WAN-VR interface ge-0/0/5.0
set routing-instances WAN-VR routing-options router-id 10.255.2.2
set routing-instances WAN-VR routing-options autonomous-system 65002
# tenant subnets: discard at preference 250, exported to the peer only by the SRG1-active node
set routing-instances WAN-VR routing-options static route 10.10.10.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.10.0/24 preference 250
set routing-instances WAN-VR routing-options static route 10.10.20.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.20.0/24 preference 250
set routing-instances WAN-VR routing-options static route 10.10.30.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.30.0/24 preference 250
set routing-instances WAN-VR routing-options static route 10.10.40.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.40.0/24 preference 250
set routing-instances WAN-VR routing-options static route 10.10.50.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.50.0/24 preference 250
set routing-instances WAN-VR routing-options static route 10.10.60.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.60.0/24 preference 250
# hub-site routes from this node's own tunnel
set routing-instances WAN-VR routing-options instance-import HUB-FROM-OVERLAY
```

Routing policy (identical on both nodes)

```
# hub-site prefixes learned over this node's own tunnel, tagged for export to the peer
set policy-options community HUB-LOCAL members 65000:1000
set policy-options policy-statement HUB-FROM-OVERLAY term hub from instance vpn_OrgOverlay
set policy-options policy-statement HUB-FROM-OVERLAY term hub from protocol bgp
set policy-options policy-statement HUB-FROM-OVERLAY term hub from route-filter 10.200.0.0/16 orlonger
set policy-options policy-statement HUB-FROM-OVERLAY term hub then community add HUB-LOCAL
set policy-options policy-statement HUB-FROM-OVERLAY term hub then accept
set policy-options policy-statement HUB-FROM-OVERLAY term end then reject
# to the peer: own default, tenant /24s only from the SRG1-active node, own hub-site routes
set policy-options policy-statement WAN-EXPORT term default from route-filter 0.0.0.0/0 exact
set policy-options policy-statement WAN-EXPORT term default then accept
set policy-options policy-statement WAN-EXPORT term lan from protocol static
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.10.0/24 exact
```

```
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.20.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.30.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.40.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.50.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.60.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from condition MNHA-ACTIVE
set policy-options policy-statement WAN-EXPORT term lan then accept
set policy-options policy-statement WAN-EXPORT term hub from community HUB-LOCAL
set policy-options policy-statement WAN-EXPORT term hub then accept
set policy-options policy-statement WAN-EXPORT term end then reject
# from the peer: hub-site routes rank below this node's own overlay routes (BGP 170)
set policy-options policy-statement ICD-IMPORT term hub from route-filter 10.200.0.0/16 orlonger
set policy-options policy-statement ICD-IMPORT term hub then preference 175
set policy-options policy-statement ICD-IMPORT term hub then accept
set policy-options policy-statement ICD-IMPORT term rest then accept
```

eBGP over ge-0/0/5 (AS 65002 to 65001, BFD 300 ms × 3) and the leak into the tenant VRs (node 2)

```
set routing-instances WAN-VR protocols bgp group ICD type external
set routing-instances WAN-VR protocols bgp group ICD family inet unicast rib-group WAN-LEAK
set routing-instances WAN-VR protocols bgp group ICD import ICD-IMPORT
set routing-instances WAN-VR protocols bgp group ICD export WAN-EXPORT
set routing-instances WAN-VR protocols bgp group ICD peer-as 65001
set routing-instances WAN-VR protocols bgp group ICD neighbor 10.255.2.1
set routing-instances WAN-VR protocols bgp group ICD bfd-liveness-detection minimum-interval 300
set routing-instances WAN-VR protocols bgp group ICD bfd-liveness-detection multiplier 3
# WAN-LEAK: every route from the peer into WAN-VR; only the peer's 0/0 into lan and GUEST-VR
set routing-options rib-groups WAN-LEAK import-rib WAN-VR.inet.0
set routing-options rib-groups WAN-LEAK import-rib lan.inet.0
set routing-options rib-groups WAN-LEAK import-rib GUEST-VR.inet.0
set routing-options rib-groups WAN-LEAK import-policy WAN-LEAK-POL
set policy-options policy-statement WAN-LEAK-POL term wan to rib WAN-VR.inet.0
set policy-options policy-statement WAN-LEAK-POL term wan then accept
set policy-options policy-statement WAN-LEAK-POL term default from route-filter 0.0.0.0/0 exact
set policy-options policy-statement WAN-LEAK-POL term default then accept
set policy-options policy-statement WAN-LEAK-POL term end then reject
```

Tenant routing instances (node 2)

```
# lan: keep Mist's overlay import, drop Mist's inet.0 default import (inet.0 holds the out-of-band management default)
set routing-instances lan routing-options instance-import vpn_OrgOverlay_bgp
delete groups top routing-instances lan routing-options instance-import wan_default
set routing-instances lan routing-options static route 0.0.0.0/0 next-table WAN-VR.inet.0
set routing-instances lan routing-options static route 0.0.0.0/0 preference 200
# GUEST-VR: GUEST and its DHCP group and pool move out of Mist's lan VR (ALQ: section 5.5)
delete groups top routing-instances lan interface ge-0/0/3.30
set routing-instances GUEST-VR instance-type virtual-router
set routing-instances GUEST-VR interface ge-0/0/3.30
delete groups top routing-instances lan system services dhcp-local-server group GUEST
delete groups top routing-instances lan access address-assignment pool GUEST
set routing-instances GUEST-VR access address-assignment pool GUEST family inet dhcp-attributes name-server 1.1.1.1
set routing-instances GUEST-VR access address-assignment pool GUEST family inet dhcp-attributes name-server 8.8.8.8
set routing-instances GUEST-VR access address-assignment pool GUEST family inet dhcp-attributes router 10.10.30.1
set routing-instances GUEST-VR access address-assignment pool GUEST family inet network 10.10.30.0/24
set routing-instances GUEST-VR access address-assignment pool GUEST family inet range GUEST high 10.10.30.200
set routing-instances GUEST-VR access address-assignment pool GUEST family inet range GUEST low 10.10.30.10
set routing-instances GUEST-VR system services dhcp-local-server group GUEST interface ge-0/0/3.30
set routing-instances GUEST-VR routing-options static route 0.0.0.0/0 next-table WAN-VR.inet.0
# no APBR: VR routing decides the exit
delete apply-groups apbr
```

Zone ICD and the tenant → ICD policies (identical on both nodes)

```
set security zones security-zone ICD interfaces ge-0/0/5.0 host-inbound-traffic system-services all
set security zones security-zone ICD interfaces ge-0/0/5.0 host-inbound-traffic protocols all
# ingress node: one tenant -> ICD policy per tenant zone
set security policies from-zone MGMT to-zone ICD policy MGMT-xnode match source-address any
set security policies from-zone MGMT to-zone ICD policy MGMT-xnode match destination-address any
set security policies from-zone MGMT to-zone ICD policy MGMT-xnode match application any
set security policies from-zone MGMT to-zone ICD policy MGMT-xnode then permit
set security policies from-zone CORP to-zone ICD policy CORP-xnode match source-address any
set security policies from-zone CORP to-zone ICD policy CORP-xnode match destination-address any
set security policies from-zone CORP to-zone ICD policy CORP-xnode match application any
set security policies from-zone CORP to-zone ICD policy CORP-xnode then permit
```

```
set security policies from-zone GUEST to-zone ICD policy GUEST-xnode match source-address any
set security policies from-zone GUEST to-zone ICD policy GUEST-xnode match destination-address any
set security policies from-zone GUEST to-zone ICD policy GUEST-xnode match application any
set security policies from-zone GUEST to-zone ICD policy GUEST-xnode then permit
set security policies from-zone MEDIA to-zone ICD policy MEDIA-xnode match source-address any
set security policies from-zone MEDIA to-zone ICD policy MEDIA-xnode match destination-address any
set security policies from-zone MEDIA to-zone ICD policy MEDIA-xnode match application any
set security policies from-zone MEDIA to-zone ICD policy MEDIA-xnode then permit
set security policies from-zone SECURITY to-zone ICD policy SECURITY-xnode match source-address any
set security policies from-zone SECURITY to-zone ICD policy SECURITY-xnode match destination-address any
set security policies from-zone SECURITY to-zone ICD policy SECURITY-xnode match application any
set security policies from-zone SECURITY to-zone ICD policy SECURITY-xnode then permit
# the GEO-BLOCK-DE lines (section 5.1) go here, above IOT-xnode, not at the end of the list
set security policies from-zone IOT to-zone ICD policy IOT-xnode match source-address any
set security policies from-zone IOT to-zone ICD policy IOT-xnode match destination-address any
set security policies from-zone IOT to-zone ICD policy IOT-xnode match application any
set security policies from-zone IOT to-zone ICD policy IOT-xnode then permit
```

ICD → local WAN and ICD → overlay (node 2, egress node)

```
# from the peer to this node's ISP (NAT: ICD_to_wan2, section 5.3)
set security policies from-zone ICD to-zone wan2 policy xnode-out match source-address any
set security policies from-zone ICD to-zone wan2 policy xnode-out match destination-address any
set security policies from-zone ICD to-zone wan2 policy xnode-out match application any
set security policies from-zone ICD to-zone wan2 policy xnode-out then permit
# from the peer into this node's tunnel to the hub
set security policies from-zone ICD to-zone vpn_OrgOverlay policy xnode-iot match source-address any
set security policies from-zone ICD to-zone vpn_OrgOverlay policy xnode-iot match destination-address any
set security policies from-zone ICD to-zone vpn_OrgOverlay policy xnode-iot match application any
set security policies from-zone ICD to-zone vpn_OrgOverlay policy xnode-iot then permit
```

Node 1 differences

```
set interfaces ge-0/0/5 unit 0 family inet address 10.255.2.1/30
set routing-instances WAN-VR routing-options router-id 10.255.2.1
set routing-instances WAN-VR routing-options autonomous-system 65001
set routing-instances WAN-VR protocols bgp group ICD peer-as 65002
set routing-instances WAN-VR protocols bgp group ICD neighbor 10.255.2.2
# lan: no preference line (ISP 1 is local)
set routing-instances lan routing-options static route 0.0.0.0/0 next-table WAN-VR.inet.0
# GUEST-VR: the local default sits behind node 2's BGP 0/0
set routing-instances GUEST-VR routing-options static route 0.0.0.0/0 next-table WAN-VR.inet.0
set routing-instances GUEST-VR routing-options static route 0.0.0.0/0 preference 200
# WAN zone is wan
set security policies from-zone ICD to-zone wan policy xnode-out match source-address any
set security policies from-zone ICD to-zone wan policy xnode-out match destination-address any
set security policies from-zone ICD to-zone wan policy xnode-out match application any
set security policies from-zone ICD to-zone wan policy xnode-out then permit
```

APBR. Mist renders `groups apbr` from the application policies. The `delete apply-groups apbr` line keeps that group from being applied; keep it in both profiles. To check, run `show configuration apply-groups` and confirm that `apbr` is not listed.

WAN-EXPORT. Term `default` exports the node's ISP default (the DHCP route in `WAN-VR`) for as long as the node holds that route. Term `lan` uses the active-signal-route condition `MNHA-ACTIVE` (section 4.4). Only the SRG1-active node advertises the tenant /24s, so the peer's `WAN-VR` sends cross-node return traffic back to the active node. Term `hub` sends only the routes from the node's own tunnel.

The cross-node policies match any source, destination and application, and zone `ICD` accepts all host-inbound services and protocols.

5.5 DHCP server with Active Leasequery (node 2 shown)

Mist renders the DHCP groups and pools from each network's DHCP settings in the Hub Profile: range .10 to .200, gateway = the SRG1 virtual IP (.1). It renders them in routing instance `lan`, inside configuration group `top`. The lines below are node 2's DHCP lines in Additional CLI.

```
# SRG1-active node answers DHCP; clients get the SRG1 virtual IP as gateway (one virtual IP per tenant network)
set chassis high-availability services-redundancy-group 1 managed-services dhcp
# SRG1 virtual IPs 1-6 (tenant gateways): section 4.4

# GUEST DHCP group and pool in GUEST-VR: section 5.4 (Tenant routing instances)

# Active Leasequery: one peer per DHCP routing instance = the other node's address in that instance
set routing-instances lan system services dhcp-local-server active-leasequery peer-address 10.10.10.2
set routing-instances lan system services dhcp-local-server active-leasequery timeout 60
set routing-instances lan system services dhcp-local-server active-leasequery idle-timeout 10
set routing-instances GUEST-VR system services dhcp-local-server active-leasequery peer-address 10.10.30.2
set routing-instances GUEST-VR system services dhcp-local-server active-leasequery timeout 60
set routing-instances GUEST-VR system services dhcp-local-server active-leasequery idle-timeout 10
set system processes dhcp-service accept-max-tcp-connections 100
set system processes dhcp-service request-max-tcp-connections 100
set system processes dhcp-service max-leasequery-reply-packet-retries 40
set system processes dhcp-service leasequery-reply-packet-retry-timer 20
set system processes dhcp-service max-leasequery-reply-packet-process 500

# lan: OFFER and ACK as broadcast (clients that do not set the DHCP broadcast flag)
set routing-instances lan system services dhcp-local-server overrides no-unicast-replies
```

Per-node value	Node 1	Node 2
Own address on MGMT / GUEST (Mist IP config)	10.10.10.2 / 10.10.30.2	10.10.10.3 / 10.10.30.3
ALQ peer in `lan`	10.10.10.3	10.10.10.2
ALQ peer in `GUEST-VR`	10.10.30.3	10.10.30.2

The `managed-services dhcp`, `no-unicast-replies` and `dhcp-service` lines are the same on both nodes. `overrides no-unicast-replies` makes the `lan` DHCP server send OFFER and ACK as broadcast, so a client that does not set the DHCP broadcast flag (for example Linux dhclient) binds. The GUEST group and pool are removed from Mist's render and re-created in `GUEST-VR` with fixed values (section 5.4). A later change to the GUEST DHCP settings in a profile does not reach `GUEST-VR`: edit the `GUEST-VR` lines in both profiles, and in a node's device-level list if it has one (section 4.2). Mist renders host-inbound `system-services dhcp` on each LAN zone, which is enough for the ALQ session (TCP/67). Check both nodes with `show dhcp server active-leasequery summary routing-instance lan` and `show dhcp server active-leasequery summary routing-instance GUEST-VR`: one peer, Connected. An unreachable peer is not listed at all.

5.6 IPsec to the hub (node 2 shown)

Mist renders the overlay from the profile settings below. The Additional CLI lines after the table are added on top of the render (node 2).

Mist setting	Value	What Mist renders
WAN port overlay path	Spoke, BFD profile broadband (section 4.3)	Certificate-based IKEv2 gateway `<wan>_to_<hub-MAC>_wan` and IPsec VPN to the hub's WAN address; multipoint `st0.0` (100.65.0.x/18); routing instance and zone `vpn_OrgOverlay` with `st0.0` and the `lo0.1` overlay loopback (100.101.0.x); iBGP group `OrgOverlay_overlay` to the hub's overlay loopback, reached over `st0.0`
Network IOT, VPN access	OrgOverlay, routed	Export policy `OrgOverlay_advertise` with the IOT subnet 10.10.60.0/24
Application policy **allow-to-hub**	IOT → hub-site networks, via OrgOverlay	IOT → `vpn_OrgOverlay` permit

```
# overlay iBGP runs the org overlay AS; the eBGP between the nodes uses WAN-VR's own AS (section 5.4)
set routing-instances vpn_OrgOverlay routing-options autonomous-system 65000

# node-local: each node negotiates its own tunnel; the wildcard matches the Mist-generated gateway name
set groups MNHA-NODE-LOCAL security ike gateway <*> node-local
set apply-groups MNHA-NODE-LOCAL

# BFD on the overlay iBGP (BGP hold time is 90 s)
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay bfd-liveness-detection minimum-interval 1000
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay bfd-liveness-detection multiplier 3

# hub prefers the SRG1-active node's IOT route (local-preference 200); the backup still advertises it at 100
set policy-options condition MNHA-ACTIVE if-route-exists address-family inet 10.255.3.1/32
set policy-options condition MNHA-ACTIVE if-route-exists address-family inet table inet.0
set policy-options policy-statement IOT-LP term active from condition MNHA-ACTIVE
set policy-options policy-statement IOT-LP term active then local-preference 200
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay export [ IOT-LP OrgOverlay_advertise ]

# DPD always-send on the Mist-generated gateway (node 2: WAN port wan2)
delete security ike gateway wan2_to_<hub-MAC>_wan dead-peer-detection optimized
set security ike gateway wan2_to_<hub-MAC>_wan dead-peer-detection always-send
```

Mist generates the instance, group and policy names from the VPN name (OrgOverlay). The overlay instance must run AS 65000, the org overlay AS that the hub uses. The eBGP between the nodes runs in WAN-VR with its own AS: 65001 on node 1, 65002 on node 2. BGP BFD (1 s × 3) comes from these lines. The BFD profile on the overlay path is a separate Mist path setting.

Master-instance AS. Each profile also sets `routing-options autonomous-system` in the master instance to the node's WAN-VR AS (65001 on node 1, 65002 on node 2). This replaces the AS 65000 that Mist renders there, so the overlay instance sets AS 65000 itself. No BGP runs in the master instance in this design.

`IOT-LP` is first in the export chain. It has no accept or reject action. It sets local-preference 200 only while the SRG1 active signal route 10.255.3.1 is in inet.0, which is true only on the active node. `OrgOverlay_advertise` then exports the IOT subnet. VR `lan` takes the hub-site routes from `vpn_OrgOverlay` through Mist's instance-import `vpn_OrgOverlay_bgp`.

Mist builds the IKE gateway name for each node from that node's WAN port name, the hub's MAC address and the hub's WAN port name: `wan2_to_<hub-MAC>_wan` on node 2 and `wan_to_<hub-MAC>_wan` on node 1. Copy it from each node's render. Check it again after a WAN port rename or a hub change. A wildcard group does not remove Mist's rendered `dead-peer-detection optimized`, so these lines name the gateway. Node 1 is identical except for the gateway name. Mist sets each node's `st0.0` address and `lo0.1` loopback.

6. Expected behavior

Each table gives what the design does in one situation and where to see it on the box. The examples use a client on the IOT VLAN.

6.1 GeoIP before and after a failover

The IOT tenant's Internet egress is node 1 (ISP 1) in both states. With node 2 active, node 2 sends the traffic across the transit link (zone ICD), so its deny counts on IOT → ICD. With node 1 active, the deny counts on IOT → wan.

Active node	Destination	Expected	Where to see it
Node 2	8.8.8.8 – **United States (US)**	**PERMITTED**	Replies; source NAT on the egress node (node 1)
Node 2	An address in GEO-DE – **Germany (DE)**	**DENIED**	GEO-BLOCK-DE hit count on node 2, IOT → ICD

Active node	Destination	Expected	Where to see it
SRG1 failover: node 2 → node 1			
Node 1	8.8.8.8 – **United States (US)**	**PERMITTED**	Replies; `01_Internet` hit count on node 1, IOT → wan
Node 1	An address in GEO-DE – **Germany (DE)**	**DENIED**	GEO-BLOCK-DE hit count on node 1, IOT → wan

```
show security policies hit-count from-zone IOT to-zone ICD          (node 2 active; excerpt)
  IOT   ICD   GEO-BLOCK-DE      3   Deny

show security policies hit-count from-zone IOT to-zone wan          (node 1 active)
  IOT   wan   GEO-BLOCK-DE      3   Deny
  IOT   wan   01_Internet     752   Permit
```

Example output; the counts grow with traffic.

6.2 Enrollment, feeds and DHCP

Check	Expected	Detail
Enrollment, both nodes	**ENROLLED**	"Device enrolled successfully!"; license type premium; device ID = chassis serial
GeoIP feed	**LOADED**	GEO-DE holds tens of thousands of IPv4 and IPv6 entries on both nodes
Backup node, including after a failover or failback	**REFRESHING**	Full GeoIP download and store; periodic feed checks about every minute; ATP Cloud Connected
DHCP lease on both nodes	**SYNCED**	For each client (match by MAC), the same IP and expiry on both nodes (section 7)
DHCP client across a manual failover / failback	**SAME IP**	The client keeps its address, in both directions

6.3 IPsec to the hub before and after a failover

Active node	Check	Expected	Where to see it
Both	IKE / IPsec / BGP to the hub	**UP**	IKEv2 UP, one IPsec SA (AES-256-GCM) and BGP Established on each node
Node 2	Ping 10.200.8.1 (hub site)	**REPLIES**	Hits on the IOT → overlay policy on node 2
SRG1 failover: node 2 → node 1			
Hub	Return route to the IOT subnet	**MOVED**	Best path changes to node 1 (local-preference 200) about 1 s after the failover
Node 1	Ping 10.200.8.1 (hub site)	**REPLIES**	Hits on the IOT → overlay policy on node 1

```
hub> show route 10.10.60.0/24 table vpn_OrgOverlay.inet.0        (after failover)
10.10.60.0/24  *[BGP/170] localpref 200, from 100.101.0.1 (node 1)  via st0.0
                [BGP/170] localpref 100, from 100.101.0.2 (node 2)  via st0.0
```

`vpn_OrgOverlay` is the routing instance Mist generates for the OrgOverlay VPN, on the hub and on both nodes.

6.4 ISP 1 failure on the active node

Node 1 is active; its tunnel uses ISP 1. ISP 1 fails and later returns. BGP over IPsec with BFD (1 s × 3) moves the hub-site traffic to node 2's tunnel.

Behavior	Expected
Hub return path	Node 1's route is withdrawn at once; the hub uses node 2's path
Node 1 outbound path	Across the transit link (ge-0/0/5), out node 2's tunnel: node 1's egress for hub-site traffic moves from st0.0 (zone `vpn_OrgOverlay`) to ge-0/0/5 (zone `ICD`)
SRG1	Stays on node 1
Interruption	**About 2.5 s in total, over the failure and the return, for new flows to the hub site**
After ISP 1 returns	Tunnel and BGP re-establish; the hub moves back to node 1 (local-preference 200) automatically

6.5 LAN link failure on the active node

Node 1 is active. Its LAN link (ge-0/0/3) goes down and later comes back; SRG1 is then moved back to node 1 by command.

Step	Event	Where to see it
Link down	Node 1 LAN link down; node 1 leaves SRG1	Node 1 `ACTIVE → INELIGIBLE, Reason: Interface monitoring down`
Takeover	Node 2 takes SRG1	Node 2 `BACKUP → ACTIVE, Reason: Peer in Ineligible state` (device event log)
Hub return path	Hub return path moves to node 2	Hub learns the IOT subnet from node 2 at local-preference 200; node 1's route is withdrawn
During the outage	Virtual IPs on node 2	The IOT virtual IP is on node 2's LAN unit
Link restored	Node 1 rejoins	Node 1 `INELIGIBLE → BACKUP, Reason: Local node recovered`; cold sync COMPLETE; SRG1 stays on node 2 (preemption disabled)
Failback	Manual SRG1 failover to node 1	Node 1 `BACKUP → ACTIVE`; hub back to node 1 (local-preference 200)

7. Verification checklist

Run each check on both nodes unless the row says otherwise.

Check	Command	Expect
MNHA state	`show chassis high-availability information`	Peer on ge-0/0/4.0: Conn State UP, Cold Sync Status COMPLETE; Peer Forwarding IP on ge-0/0/4.0 with Peer ICD Conn State UP; SRG1 ACTIVE on one node, BACKUP on the other; Preemption DISABLED; Failure Events: NONE
LAN monitoring	`show chassis high-availability services-redundancy-group 1`	Interface Monitoring: `ge-0/0/3` State Up, on both nodes
MNHA config	`show configuration chassis high-availability | display inheritance no-comments | display set | no-more`	The `chassis high-availability` lines of the node's own profile (section 4.4), and no others. A `chassis high-availability` line removed from the profile stays on the box until it is deleted there by CLI
Session sync	`show chassis high-availability data-plane statistics`	Session create: RTOs received on the backup rise with RTOs sent on the active node. Compare the change over a minute, not the totals
Sessions on both nodes	`show security flow session summary`	Similar valid-session count on both nodes
Transit link BGP	`show bgp summary instance WAN-VR`	Group ICD peer Established: 10.255.2.2 on node 1, 10.255.2.1 on node 2
Transit link routes	`show route receive-protocol bgp <peer> table WAN-VR.inet.0 | no-more`	From the peer (10.255.2.2 on node 1, 10.255.2.1 on node 2): 0/0 while the peer holds its ISP DHCP default, and the hub-site prefixes while the peer's tunnel is up. The six tenant /24s arrive only on the backup node, from the SRG1-active peer
Tenant egress	`show route 0.0.0.0/0 exact table lan.inet.0 | no-more` `show route 0.0.0.0/0 exact table GUEST-VR.inet.0 | no-more`	With both ISPs up. lan: Static (to table WAN-VR.inet.0) active on node 1; BGP via 10.255.2.1 active on node 2. GUEST-VR: BGP via 10.255.2.2 active on node 1; Static (to table WAN-VR.inet.0) active on node 2
APBR removed	`show configuration apply-groups`	`apbr` not listed
BFD	`show bfd session`	All Up. Transit link peer on ge-0/0/5.0: 300 ms × 3. Hub overlay neighbor: 1000 ms × 3. MNHA ICL and forwarding IP (10.255.0.x, 10.255.1.x): 1000 ms × 3
IPsec	`show security ike security-associations` `show security ipsec security-associations`	UP on both nodes
Overlay BGP	`show bgp summary instance vpn_OrgOverlay`	Hub neighbor Established on both nodes

Check	Command	Expect
Signal route on the active node only	`show route 10.255.3.1/32 exact`	Active node: 10.255.3.1/32 present (the route the MNHA-ACTIVE condition tests). Backup: no 10.255.3.1/32 (it holds 10.255.3.2/32)
Hub return path	hub: `show route 10.10.60.0/24 table vpn_OrgOverlay.inet.0`	Best path = active node (local-preference 200); backup node at 100
Cloud connection	`show services advanced-anti-malware status`	Connected on both nodes
Feed refresh	`show services security-intelligence update status`	succeeded; time advances on both nodes
GeoIP object	`show security dynamic-address summary`	GEO-DE IPv4/IPv6 entries > 0
GeoIP policy order	`show security policies from-zone IOT to-zone wan` (node 1) `show security policies from-zone IOT to-zone wan2` (node 2) `show security policies from-zone IOT to-zone ICD`	GEO-BLOCK-DE listed first (sequence 1), above 01_Internet or IOT-xnode. Recheck after any Mist application-policy change
Policy enforced	`show security policies hit-count`	GEO-BLOCK-DE increases on the active node: IOT → wan on node 1; IOT → ICD on node 2 (its lan tenants exit ISP 1 through node 1)
NAT	`show security nat source rule all`	Translation hits increase on the egress node: `<tenant>_to_wan` (node 1) or `<tenant>_to_wan2` (node 2) for local traffic; `ICD_to_wan` / `ICD_to_wan2` for traffic from the peer
DHCP sync	`show dhcp server active-leasequery summary routing-instance <ri>`	Peer Connected for lan and for GUEST-VR, on both nodes. A missing peer line means the ALQ session to that peer is down (an unreachable peer is not listed)
DHCP bindings	`show dhcp server binding routing-instance <ri>`	For each client (match by MAC), the same IP address and lease expiry on both nodes. A matching binding count alone does not show this
Mist commit	`show system commit`	A new commit by Mist after each profile change on that node. Check this before acting on the portal status: a commit shown as FAILED can later show COMMITTED, for example after a retry
Manual failover / failback	`request chassis high-availability failover services-redundancy-group 1 peer-id <target>` (on the active node)	Run when the target node shows Cold Sync COMPLETE, ALQ peer Connected, the same IP for each client on both nodes, and overlay BGP and BFD Up. Preemption is disabled, so failback is manual

`vpn_OrgOverlay` and `OrgOverlay_overlay` are the names Mist generates for the overlay named OrgOverlay. WAN-VR holds the WAN and the transit link (section 5.4). In `show chassis high-availability information`, Peer ICD is the MNHA forwarding path on the ICL (ge-0/0/4.0). The transit link on ge-0/0/5 does not appear there; check it with the transit link BGP and BFD rows.

8. Deployment order

The configuration blocks of sections 4 and 5, in dependency order.

0. **Prerequisites.** In Mist, create the OrgOverlay VPN, the org networks MGMT, CORP, GUEST, MEDIA, SECURITY and IOT, and the HUB-SITE-NETS application that allow-to-hub uses. Build the vSRX hub's Hub Profile (hub role, static WAN; section 4.5), assign it to the hub, and confirm that the hub is connected. On the LAN switch, carry VLAN 10 (MGMT) untagged and VLANs 20, 30, 40, 50 and 60 tagged on the trunk to ge-0/0/3 of each node. The virtual IPs and both ALQ peers need these VLANs on both nodes (section 3.6).
1. In Mist, create one Hub Profile per SRX400 node (not a WAN Edge Template) and assign each to its device (section 4.2). Keep the Additional CLI in the profile. A device-level Additional CLI list replaces the profile's list; if a node has one, edit both. After every change, confirm the Mist commit on the node (`show system commit`).
2. Set the native fields in each profile: WAN port, LAN trunk with the node's own address on each tenant network, DHCP pools, IOT VPN access, the overlay spoke path, the traffic steering paths wan and OrgOverlay, and the application policies (section 4.3).
3. Add MNHA to each profile's Additional CLI: the ICL /30 on ge-0/0/4, the lo0.0 forwarding address with a static route to the peer's, MNHA forwarding on ge-0/0/4.0, zone HA, and SRG1 with virtual IPs, signal routes and `monitor interface ge-0/0/3` (section 4.4). Check the MNHA state and LAN monitoring rows (section 7).
4. Add the cross-node routing and policy (section 5.4): WAN-VR holding the WAN port and ge-0/0/5, the inet.0 management default, eBGP with BFD on the transit link (WAN-EXPORT, ICD-IMPORT, rib-group WAN-LEAK), the VR lan and GUEST-VR defaults, `delete apply-groups apbr`, zone ICD with its cross-node policies, and ICD source NAT (section 5.3). Check the transit link BGP and BFD rows.
5. Add the overlay lines (section 5.6): `vpn_OrgOverlay` autonomous-system 65000, the master-instance `routing-options autonomous-system` (65001 on node 1, 65002 on node 2), the `MNHA-NODE-LOCAL` group, BGP BFD 1000 ms × 3, condition MNHA-ACTIVE with IOT-LP first in the export chain, and DPD always-send on the Mist-generated gateway (`wan_to_<hub-MAC>_wan` on node 1, `wan2_to_<hub-MAC>_wan` on node 2). Add the hub's lines to its profile (section 4.5). Check the IPsec, overlay BGP, signal route and hub return path rows.
6. Add `managed-services dhcp`, broadcast replies in lan (`overrides no-unicast-replies`) and an Active Leasequery peer in each DHCP routing instance, lan and GUEST-VR (section 5.5). Check the DHCP rows.
7. Confirm each chassis serial is entitled for ATP Cloud. Give each node a path to ATP Cloud and DNS. In this design that path is the out-of-band management network, through the inet.0 default (section 3.5).
8. Enroll each node with the ATP Cloud portal command (section 3.1). Then add the ATP Cloud / SecIntel lines to the profile (section 5.2); they reference the certificates and CA profiles that enrollment creates. Check the cloud connection and feed refresh rows.
9. Add the GeoIP object and the GEO-BLOCK-DE policies with `set` and `delete` lines only; Mist Additional CLI also accepts `#` comment lines, but not `insert` (section 5.1). In the Additional CLI list, place the IOT → ICD GEO-BLOCK-DE lines above the IOT-xnode lines from step 4; do not append them at the end of the list. Check the GeoIP object and policy order rows: GEO-BLOCK-DE is sequence 1 on IOT → ICD and on IOT → wan (node 1) or wan2 (node 2).
10. From a tenant host, check a permitted (US) destination, a denied (DE) destination and the hub site (10.200.8.1) against section 6. Fail over with the manual failover row, check again, then fail back.

11. Add the IPv6 block (section 9) as the last block of node 1's Additional CLI, below the `# [after enrollment]` blocks; replace the delegated /60 first (section 9.1). Check the rows of section 9.3.

9. IPv6 on the ISP 1 tenants (node 1)

IPv6 runs on node 1 only, toward ISP 1, for the five tenants that use ISP 1: MGMT, CORP, MEDIA, SECURITY and IOT. GUEST (ISP 2) and node 2 stay IPv4-only. Every line is Additional CLI in node 1's Hub Profile, as the last block of its list, below the four `# [after enrollment]` blocks.

9.1 How it works

- **WAN: DHCPv6 in `WAN-VR`** (the WAN placement of section 5.4). `ge-0/0/2.0` runs a stateful DHCPv6 client that requests an address (IA_NA) and a delegated prefix (IA_PD, preferred length /60), with a DUID-LL client identifier. The IPv6 default (::/0 toward ISP 1, learned from its router advertisements) is installed in `WAN-VR.inet6.0` only when the WAN unit is listed under `router-advertisement`, so the block lists `ge-0/0/2.0` there; node 1 then also sends router advertisements on the WAN unit. Zone `wan` accepts the dhcpv6 host-inbound service. The block has no `security forwarding-options family inet6` line: on the SRX400, `show security flow status` reports *Inet6 forwarding mode: flow based* without it.
- **LAN: one static /64 per tenant, from the delegated /60.** Node 1 has address ::2 in each /64 and sends router advertisements with that prefix on the tenant's unit of `ge-0/0/3`. The last hex digit of the fourth group (the /64's subnet ID inside the /60) is the tenant's VLAN divided by 10.

Tenant	Unit	/64 (node 1 address ::2)
MGMT	`ge-0/0/3.0`	2001:db8:2000:1::/64
CORP	`ge-0/0/3.20`	2001:db8:2000:2::/64
MEDIA	`ge-0/0/3.40`	2001:db8:2000:4::/64
SECURITY	`ge-0/0/3.50`	2001:db8:2000:5::/64
IOT	`ge-0/0/3.60`	2001:db8:2000:6::/64

- **Routing.** `lan.inet6.0` has a static default `next-table WAN-VR.inet6.0`, one way only. The delegated /60 is a discard route in `WAN-VR.inet6.0`; `WAN-VR` holds no route to the five /64s. Return traffic to a LAN host follows its firewall session.
- **Policy and NAT.** Mist renders each tenant's `01_Internet` policy to zone `wan` with the tenant's IPv4 network as its source. The block adds the source `any-ipv6` to that policy for the five tenants. IPv6 is not translated: the tenant source NAT rules match 0.0.0.0/0. In node 1's profile the IPv6 block is the last block, below the `# [after enrollment]` block that deletes and re-sets IOT's `01_Internet` (section 5.1).
- **Node 1 only.** Node 2 has no IPv6 lines, and the /64 addresses are node 1's own, not SRG1 virtual IPs. IPv6 stays on node 1 when SRG1 moves to node 2. There is no IPv6 path through node 2 or ISP 2.
- **Fixed prefix.** 2001:db8:2000::/60 is an example delegated /60 (a documentation prefix). Replace it with your delegated /60 in the 11 lines that carry it: five addresses, five router-advertisement prefixes and the discard route. Keep the last digit of each /64's fourth group: with 2001:db8:abcd:1230::/60, CORP is 2001:db8:abcd:1232::/64 and node 1 is 2001:db8:abcd:1232::2. The lines do not follow the delegation: a different /60 from the ISP means changing them.

9.2 Configuration (node 1 Additional CLI)

```
# WAN ge-0/0/2.0 (in WAN-VR): DHCPv6 address (IA_NA) + delegated /60 (IA_PD)
set interfaces ge-0/0/2 unit 0 family inet6 dhcpv6-client client-type stateful
set interfaces ge-0/0/2 unit 0 family inet6 dhcpv6-client client-ia-type ia-na
set interfaces ge-0/0/2 unit 0 family inet6 dhcpv6-client client-ia-type ia-pd
set interfaces ge-0/0/2 unit 0 family inet6 dhcpv6-client prefix-delegating preferred-prefix-length 60
set interfaces ge-0/0/2 unit 0 family inet6 dhcpv6-client client-identifier duid-type duid-ll

# zone wan accepts DHCPv6
set security zones security-zone wan host-inbound-traffic system-services dhcpv6

# router-advertisement on the WAN unit (the IPv6 default needs it, 9.1)
set protocols router-advertisement interface ge-0/0/2.0

# LAN: node 1 address ::2 in one /64 per tenant (MGMT .0, CORP .20, MEDIA .40, SECURITY .50, IOT .60)
set interfaces ge-0/0/3 unit 0 family inet6 address 2001:db8:2000:1::2/64
set interfaces ge-0/0/3 unit 20 family inet6 address 2001:db8:2000:2::2/64
set interfaces ge-0/0/3 unit 40 family inet6 address 2001:db8:2000:4::2/64
set interfaces ge-0/0/3 unit 50 family inet6 address 2001:db8:2000:5::2/64
set interfaces ge-0/0/3 unit 60 family inet6 address 2001:db8:2000:6::2/64

# router advertisements: the tenant /64 on each unit
set protocols router-advertisement interface ge-0/0/3.0 prefix 2001:db8:2000:1::/64
set protocols router-advertisement interface ge-0/0/3.20 prefix 2001:db8:2000:2::/64
set protocols router-advertisement interface ge-0/0/3.40 prefix 2001:db8:2000:4::/64
set protocols router-advertisement interface ge-0/0/3.50 prefix 2001:db8:2000:5::/64
set protocols router-advertisement interface ge-0/0/3.60 prefix 2001:db8:2000:6::/64

# the delegated /60 discards in WAN-VR; lan sends its IPv6 default to WAN-VR
set routing-instances WAN-VR routing-options rib WAN-VR.inet6.0 static route 2001:db8:2000::/60 discard
set routing-instances lan routing-options rib lan.inet6.0 static route ::/0 next-table WAN-VR.inet6.0

# Mist's 01_Internet (tenant -> wan) also matches IPv6 sources; no IPv6 NAT
set security policies from-zone MGMT to-zone wan policy 01_Internet match source-address any-ipv6
set security policies from-zone CORP to-zone wan policy 01_Internet match source-address any-ipv6
set security policies from-zone MEDIA to-zone wan policy 01_Internet match source-address any-ipv6
set security policies from-zone SECURITY to-zone wan policy 01_Internet match source-address any-ipv6
set security policies from-zone IOT to-zone wan policy 01_Internet match source-address any-ipv6
```

The same lines, in Junos order, are in `junos/srx400-node1.set` (section 10). Node 2's profile has no IPv6 lines.

9.3 Expected behavior

Tenant hosts use node 1 as their IPv6 gateway.

Check	Expected	Detail
DHCPv6 on `ge-0/0/2.0` in `WAN-VR`	**BOUND**	Address (IA_NA) and delegated prefix 2001:db8:2000::/60 (IA_PD)
IPv6 default in `WAN-VR`	**INSTALLED**	::/0 from ISP 1's router advertisements
Five tenants through ISP 1, SRG1 active on node 1	**REPLIES**	MGMT, CORP, MEDIA, SECURITY, IOT
Five tenants through ISP 1, SRG1 active on node 2	**REPLIES**	IPv6 stays on node 1 and ISP 1
Firewall session (CORP shown)	**PERMITTED**	Policy `01_Internet`, in `ge-0/0/3.20`, out `ge-0/0/2.0`, no NAT
Router advertisements on the five LAN units	**SENT**	Each with its tenant /64 (9.1)
GUEST and node 2	**IPv4 ONLY**	No IPv6 address on GUEST; no IPv6 lines and no router advertisements on node 2

Check	Expected	Detail
Node 1 reboot	**BOUND**	DHCPv6 binds again on `ge-0/0/2.0` (DUID-LL client identifier)
IPv4 alongside IPv6	**UNCHANGED**	CORP exits ISP 1; GUEST exits ISP 2 across `ge-0/0/5`

10. Reference configurations

Generic configurations of this design (sections 4 and 5), with node 1's IPv6 lines (section 9), are published with this paper at github.com/SandmanJnpr/srx400-mnha-mist-branch. They use this paper's names and addresses; the Hub Profiles are named SRX400-1 (node 1), SRX400-2 (node 2) and HUB (vSRX hub). Installation values, such as the management addresses, the hub's MAC address in the IKE gateway names and the ATP Cloud region, are placeholders in angle brackets, and node 1's delegated IPv6 /60 is an example (9.1). This is a reference by the author, not an official Juniper or HPE configuration.

File	Contents
`README.md`	Links to the paper and the configuration files
`docs/SRX400-MNHA-Secure-Branch-Design.pdf`	This paper
`docs/topology.png`	The topology diagram of section 1
`mist/srx400-node1.json` `mist/srx400-node2.json`	Hub Profiles of node 1 and node 2: native settings and Additional CLI (node 1: with the IPv6 block of section 9)
`mist/vsrx-hub.json`	Hub Profile of the vSRX hub, which the node profiles' overlay needs
`junos/srx400-node1.set` `junos/srx400-node2.set`	Each node's configuration in display set form with the generic values, in Junos order; adoption, access and box-local lines, DHCP reservations and secrets left out. Node 1's file includes the IPv6 lines of section 9

Installation-specific values (organization name, MAC and management addresses, device status) are covered in Figures 4-1 to 4-5.